

Imperial College London
Department of Earth Science and Engineering
MSc in Applied Computational Science and Engineering

Independent Research Project
Project Plan

Parameter Sensitivity Analysis and Machine Learning Prediction of Tidal Disruption Outcomes in SPH Simulations of Martian Moon Formation

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Abstract

This project investigates how orbital, physical, and numerical parameters influence tidal-disruption outcomes in SPH simulations of asteroid encounters with Mars. The project develops a pipeline to extract FoF fragment properties, connect them with physical snapshot data, and derive bound-aware outcomes such as bound mass fraction, bound fragment count, and largest bound fragment mass. Exploratory analysis will identify dominant physical and numerical controls, while machine-learning models will test how predictable these outcomes are from simulation metadata. Modelling will use explicit baselines, regularised linear/logistic models, and tree-based ensembles, with leakage-aware validation and limited tuning where justified. The expected deliverables are structured datasets, sensitivity results, predictive-model benchmarks, and an interpretation of parameter regimes relevant to debris retention around Mars.

1. Introduction

1.1. Problem Description and Research Questions

Mars's moons, Phobos and Deimos, have an uncertain origin. Their asteroid-like appearance suggests capture, but their low-inclination, near-circular orbits are difficult to explain by simple capture alone. Impact-generated debris disks provide another explanation, but require specific impact conditions and subsequent disk evolution. A recent alternative is disruptive partial capture, in which an asteroid passing close to Mars is tidally disrupted and some fragments are retained in Mars orbit (Kegerreis et al., 2024).

This IRP asks: how do orbital and physical parameters influence tidal-disruption outcomes; which parameters most strongly control fragment formation and bound debris mass; and can machine-learning models predict disruption outcomes across the sampled parameter space? The computational challenge is to extract, organise, and model an existing SPH simulation campaign, rather than to run new simulations from scratch.

1.2. Significance

This work connects Martian moon formation with a practical computational-science problem: turning raw particle simulation outputs into structured datasets for analysis and prediction. Identifying encounter conditions that favour retained debris helps evaluate disruptive partial capture as a plausible pathway for Phobos and Deimos formation.

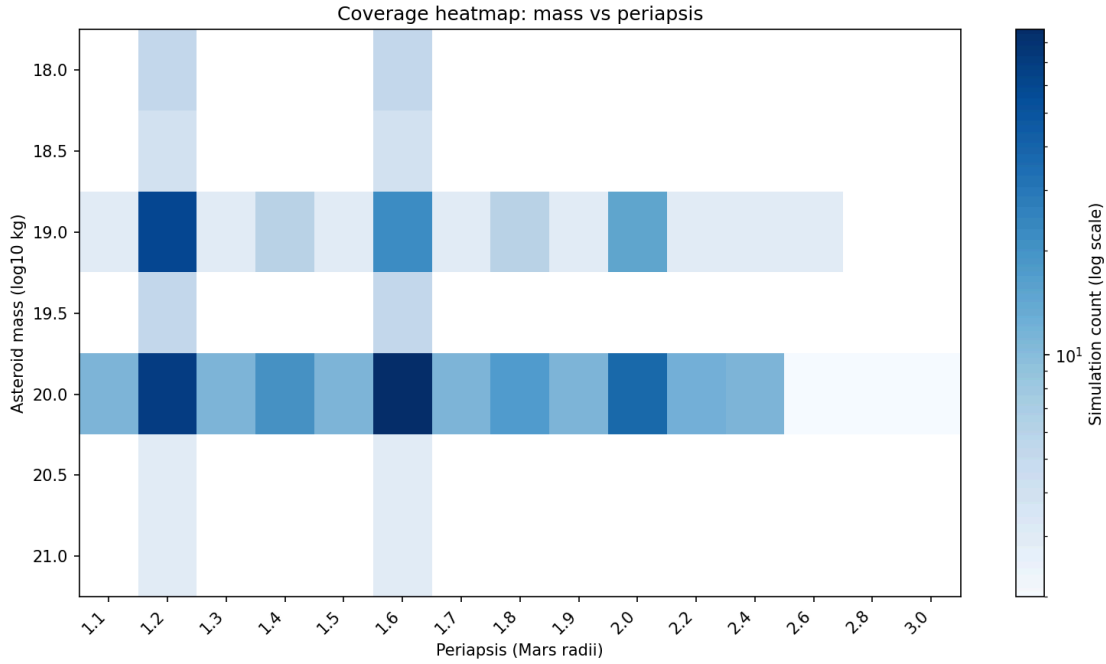


Figure 1. Existing simulation coverage across asteroid mass and periapsis. Uneven sampling motivates both full-dataset analysis and controlled subsets.

1.3. Literature Review

Several formation mechanisms have been proposed for Phobos and Deimos. Giant-impact studies show that impacts on Mars can create debris disks massive enough to form small moons (Citron et al., 2015; Canup & Salmon, 2018), while Rosenblatt et al. (2016) argued that transient inner moons could stir an outer disk and support accretion of small outer moons. Other work proposes that the moons may be remnants of a disrupted common progenitor (Bagheri et al., 2021). These studies demonstrate that the origin problem is active, but they also show that debris production, capture, and later orbital evolution remain difficult to connect.

Kegerreis et al. (2024) provide the most direct motivation for this project by modelling disruptive partial capture of an asteroid near Mars. Their work suggests that tidal disruption can capture a non-negligible fraction of asteroid material over a range of periapsis distances, velocities, masses, spins, and orientations. This IRP builds on that scenario by analysing the SPH disruption outputs themselves: instead of asking only whether capture is possible, it asks which input parameters control fragment and bound-debris outcomes.

SPH is appropriate because it is a mesh-free Lagrangian method widely used for astrophysical, planetary, and impact-like simulations, although its numerical choices and post-processing steps require careful interpretation (Price, 2012). However, fragment identification introduces post-processing uncertainty. FoF algorithms group particles using a linking length, and previous FoF studies show that inferred structures can depend strongly on this parameter (More et al., 2011). Therefore, FoF fragment counts are useful proxies, but they should not be treated as final physical moon-forming outcomes.

Machine learning is increasingly used to extract patterns from simulation outputs and identify controls in complex physical systems (Brunton et al., 2020). Related astrophysical simulation projects, such as CAMELS, also use ML workflows to map input parameters to simulation-derived observables (Villaescusa-Navarro et al., 2021). Here, ML is diagnostic rather than a replacement for physical analysis: simple reference models and tree-based ensembles, including random forests and gradient boosting, will test predictability, feature importance, residual structure, and sensitivity to post-processing choices (Breiman, 2001; Friedman, 2001).

1.4 Objectives

Table 1. Objectives and measurable success criteria

Objective	Success criterion
Build a fragment and bound-debris extraction pipeline	Generate structured tabular datasets for initial conditions, FoF-derived outcomes, and bound/unbound metrics for the available successful simulations
Quantify parameter sensitivity	Identify the dominant physical and numerical controls using EDA on both full and controlled subsets
Develop and validate predictive models	Train explicit baseline model families, including dummy predictors, regularised linear/logistic models, random forests, and gradient boosting models; apply limited tree-model tuning only where baseline results show learnable structure
Distinguish physics from post-processing	Compare model behaviour with and without FoF linking length, and use grouped validation where alternative FoF outputs share the same physical snapshot
Deliver a defensible interpretation	Produce a final report explaining results, limitations, and implications for the disruptive-capture scenario

2. Methodology

2.1 Data and Simulation Outputs

The project uses existing SPH simulations of asteroid encounters with Mars, with output data stored in HDF5 format. FoF outputs provide group identities and fragment-size proxies, while physical snapshots provide particle coordinates, velocities, and masses. These sources are joined into simulation-level and fragment-level tables.

2.2 Fragment and Bound-Debris Extraction

The extractor will compute fragment statistics and classify material as bound or unbound relative to Mars using specific orbital energy. The main targets will be bound mass fraction, bound fragment count, and largest bound fragment mass. These are closer to the formation question than FoF counts alone, although bound material is only a necessary, not sufficient, condition for a long-term moon-forming disk.

2.3 Parameter Sensitivity Analysis

Exploratory analysis will compare the full dataset with controlled subsets that fix timestep, dominant resolution, and FoF linking length. This is needed because the data is unevenly sampled and because numerical or post-processing choices can obscure physical trends.

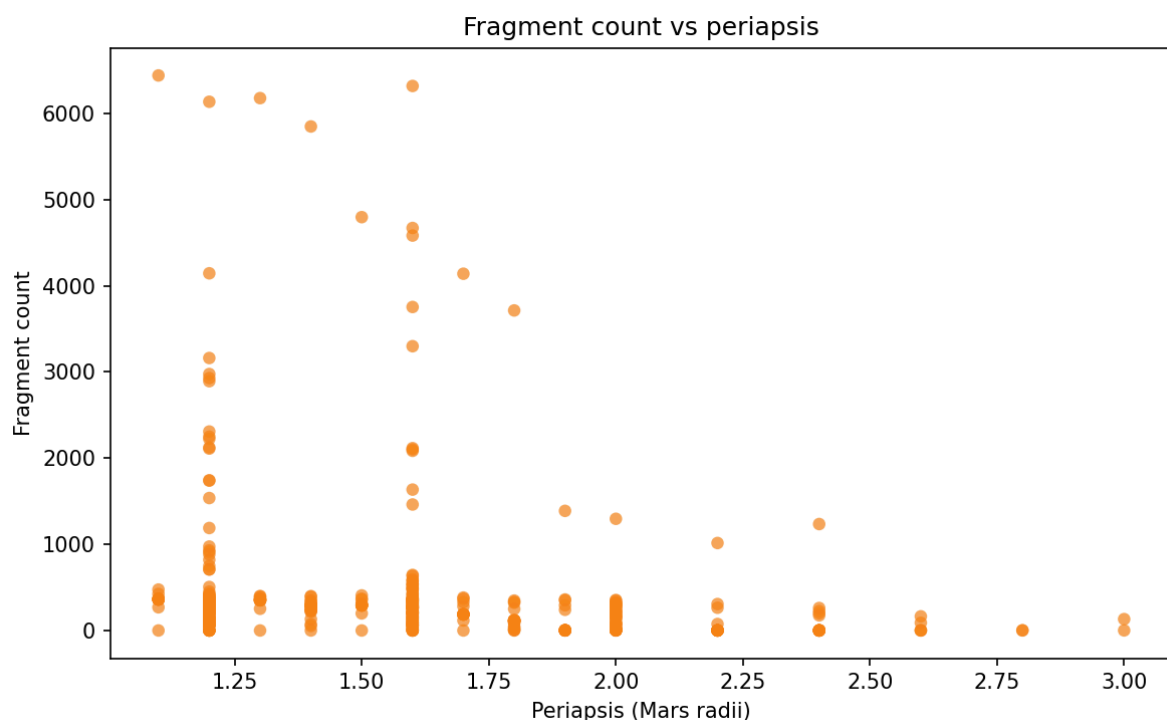


Figure 2. Current FoF trend between periapsis and fragment count. Closer encounters tend to produce more detected fragments.

2.4 Machine Learning Prediction and Validation

ML models will use one row per simulation to test whether disruption outcomes are predictable from simulation metadata. Input features will include asteroid mass, resolution, periapsis distance, encounter velocity, spin period, spin-axis category, timestep, and FoF linking length. Models will be trained both with and without FoF linking length to separate physical controls from post-processing sensitivity.

The modelling will first compare fixed baseline families: dummy predictors, ridge/logistic regression, random forests, and gradient boosting models. Regression targets will include fragment counts, largest fragment mass, bound mass fraction, bound fragment count, and largest bound fragment mass. Classification targets will include whether any bound mass is retained and whether bound mass fraction exceeds a threshold such as 0.1.

If baseline results show learnable structure, limited tuning will be applied to tree models only, using cross-validated searches over number of trees, maximum depth, minimum leaf size, learning rate, and subsampling. Grouped folds by physical file will prevent leakage between alternative FoF linking-length versions of the same simulation. Performance will be reported using MAE, RMSE, and R^2 for regression; balanced accuracy, F1, and ROC-AUC for classification; plus residual, calibration, and feature-importance diagnostics.

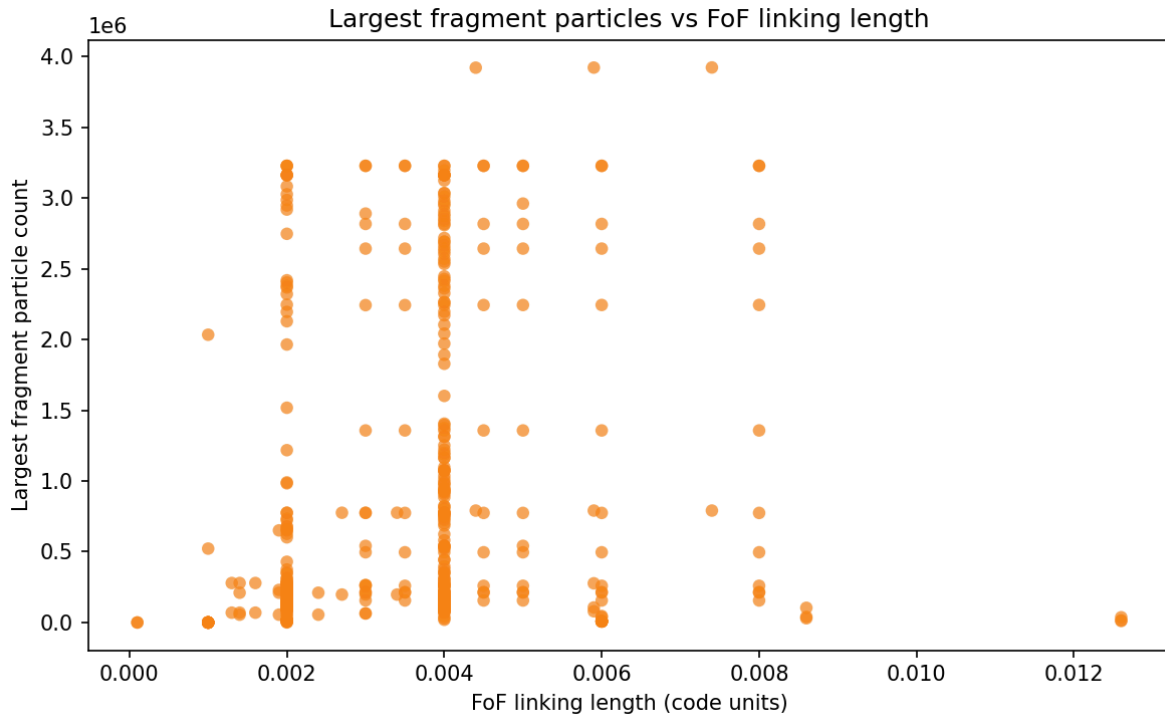


Figure 3. Largest detected fragment particle count as a function of FoF linking length. The scatter suggests that FoF linking length may affect fragment measurements, but it is not the only controlling factor. This motivates checking post-processing effects by comparing model results with and without FoF linking length.

2.5 Limitations and Alternative Approaches

Running a new simulation campaign would give cleaner coverage, but would exceed the three-month IRP scope. A purely descriptive project would be safer but would miss the chance to test predictability. The chosen approach balances feasibility, physical interpretation, and computational analysis.

3. Progress to Date

Preliminary work has produced FoF extraction for 489 simulation outputs, initial EDA, and baseline ML diagnostics. The current bound-aware pipeline also matches FoF fragment outputs to physical snapshots and labels fragments as bound or unbound using specific orbital energy relative to Mars. The first bound-aware EDA contains 189,664 fragment rows, 407 successful run rows, and 39,479 bound fragments, with energy sign agreeing with the bound label for all fragment rows. Benchmark ML results are already informative: the best current classifier for retaining any bound mass reaches balanced accuracy of 0.927, and the best bound-mass-fraction regressor reaches R^2 of 0.897 when FoF linking length is included. These remain benchmark results until residual checks, controlled-subset comparisons, and limited tuning are completed.

4. Future Plan, Feasibility, and Risks

Table 2. Planned timeline for the remaining IRP period

Period	Main tasks	Milestones
June 2026	Finalise bound-aware extraction/EDA, update project plan, and verify leakage-safe validation design	Bound-aware dataset, EDA summary, and benchmark model plan
July 2026	Run full model comparison, residual diagnostics, controlled-subset checks, and limited tree-model tuning where justified	Defend sensitivity results and predictive-model benchmarks complete
August 2026	Interpret physical implications, refine figures, separate FoF proxy conclusions from bound-aware conclusions, and write final report	Final narrative, limitations, and conclusions complete

Main risks are manageable. If further HPC or data-access issues arise, analysis can continue from compact CSV outputs already generated by the extraction pipeline. Sparse or noisy targets will be handled using both regression and classification targets, plus full-dataset and controlled-subset analysis. If ML adds limited value for some outcomes, it will remain a diagnostic benchmark. If FoF choices dominate, results will be compared with and without linking length, and bound mass will be presented as a proxy rather than proof of long-term moon-forming disk material.

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